

School of Computer Science and Engineering

Internet of Things Standard Operating Procedure

Effective:

1. Objective: This course will introduce the fundamental concepts of the Internet of Things (IoT), including its architecture, ecosystem, and real-world applications. It will help the students to develop the ability to build, test, and evaluate IoT applications involving data acquisition, cloud integration, and basic analytics.

2. Scope: This policy applies to 2024, 2025 BTech (Hons.) Batch students enrolled at the School of Computer Science & Engineering, RV University, Bangalore.

3. Credits and Hours of Effort

- Mandatory 3-credit course, (LTP::0:2:2) as per RVU curriculum & UGC guidelines.

4. Students has to complete the following components for the course:

1. MOOCs Course: NPTEL course on [Introduction To Internet Of Things](#) by Prof. Sudip Misra, IIT Kharagpur
2. Mini Project : Develop an IOT based project using microcontroller, sensors, etc. under the guidance of the internal Guide

5. Mini Project

The IoT mini project should aim to solve a real-world problem or improve an existing process using IoT technologies. The solution should involve data sensing, transmission, and decision-making based on data analysis or control logic.

a) Team Composition

- Each team should consist of **4 members of the same section**.
- Teams must designate a **team leader** for coordination and communication with mentors or faculty.

b) Project Selection

- Projects should be **innovative**, feasible within the timeline.
- Probable topics such as:

- Smart Home Automation
- IoT-based Weather Monitoring
- Smart Irrigation System
- Health Monitoring System
- Industrial Equipment Monitoring
- Smart Parking System

c) Components Required

The project can include:

- Sensors (e.g., temperature, humidity, gas, motion, etc.)
- Microcontroller (e.g., Arduino, ESP32, NodeMCU, etc.)
- Connectivity Module (e.g., Wi-Fi, Bluetooth, LoRa)
- Actuators (e.g., relays, motors, LEDs)

d) Project Documentation

The project report (Format will shared in due course) must include:

- Title Page
- Abstract (Problem statement + proposed solution)
- Introduction
- Literature Review
- Hardware and Software Requirements
- System Architecture / Block Diagram
- Working Principle
- Circuit Diagrams and Screenshots
- Code Explanation
- Testing and Results
- Conclusion and Future Scope
- References

e) Timeline

- Project Proposal Submission: [within Week 3]
- Component Finalization: [Week 4-5]
- Development & Testing: [Week 10-12]
- Final Presentation and Demonstration: [Week 14]

6. MOOCs Course:

Students must register a NPTEL course on “Introduction To Internet Of Things” by Prof. Sudip Misra, IIT Kharagpur. This course will introduce the fundamental concepts of the Internet of Things (IoT), including its architecture, ecosystem, and real-world applications.

- a) Session 1: Introduction to IOT and Development Platforms
 - [INTRODUCTION TO IOT- PART-I](#)
 - [INTRODUCTION TO IOT- PART-II](#)
 - [INTRODUCTION TO ARDUINO-I](#)
 - [INTRODUCTION TO ARDUINO-II](#)
 - [CASE STUDY: HEALTH CARE](#)
- b) Session 2 : Sensors and Actuators – Interfacing and Control
 - [SENSING](#)
 - [ACTUATION](#)
 - [INTEGRATION OF SENSOR AND ACTUATORS WITH ARDUINO-I](#)
 - [CASE STUDY: AGRICULTURE](#)
- c) Session 3: Networking and IoT Protocols
 - [BASICS OF IOT NETWORKING- PART- I](#)
 - [BASICS OF IOT NETWORKING-PART II](#)
 - [BASICS OF IOT NETWORKING-PART- III](#)
- d) Session 4: Wireless Sensors Network Introduction
 - [SENSOR NETWORKS-I](#)
 - [SENSOR NETWORKS-II](#)
 - [SENSOR NETWORKS-III](#)
 - [SENSOR NETWORKS-IV](#)
 - [SENSOR NETWORKS-V](#)
- e) Session 5: Sensor Network Using Cloud
 - [SENSOR-CLOUD- I](#)
 - [SENSOR-CLOUD- II](#)
- f) Session 6: IOT Applications
 - [SMART CITIES AND SMART HOMES- I](#)
 - [SMART CITIES AND SMART HOMES- II](#)
 - [SMART CITIES AND SMART HOMES- III](#)

Note: The NPTEL course covers a wide range of additional topics beyond the content above. Students who are interested are encouraged to explore the extended material for deeper learning and to make the most of the course. Assignment submissions must be completed on the NPTEL portal within the

specified deadlines to be eligible for the proctored exam. Students may choose to appear for the proctored exam on their own initiative and earn a certificate. **Holding an NPTEL certificate can add significant value to your resume.**

7. Continuous Internal Evaluation (CIE) and Semester End Examination (SEE)

<u>Internal Assessment Plan: 70 Marks</u>			
Sl#	Marks	Type of Assessment	Timeline
CIE 1 (20 Marks)			
1	20	1. Project review 1: Proposal Submission and Component Finalization (10 marks) 2. Quiz (10 marks)	Week 4
CIE 2 (25 Marks)			
2	25	Midterm exam/CIE-02 based on MOOCs content	Week 10
CIE 3 (25 Marks)			
3	25	Project review 2: Development & Testing	Week 13

Project Review 1 and 2:

- The Internal faculty allotted will award **Project Review 1 and 2** of CIE 1 and CIE 3 marks respectively based on the rubrics drafted.
- CIE 1 Quiz (10 marks) will be conducted from session 1 and 2 of the NPTEL course.
- For CIE 1 Quiz, 20 Questions will be provided with time duration of 15 mins.

CIE2/ Midterm exam:

- Pen and Paper based exam
- Based on all above mentioned sessions of NPTEL course
- Question Paper Pattern

<u>Mid Sem Assessment Pattern: 25 Marks</u>	
Sl#	Content
Part A- 5 Mark, (Minimum 1 marks, Max 2 marks , Max 5 Questions)	
Part B Total marks - 20,(Compulsory 4 questions of 5 marks each, there can be max 3 sub division)	

8. Rubrics

Evaluation Rubrics:(CIE-70 marks):

- **CIE 1 - Project review 1: Proposal Submission and Component Finalization & Procurement (10 Marks)**

Includes:

1. Problem Statement(2)
2. Introduction and Literature Survey(2)
3. Methodology to be used(3): Hardware and Software Requirements, Block Diagram, Working Principle, Circuit Diagrams and Screenshots
4. Presentation Skills(1)
5. Viva(2)

Criteria	Max Marks	Evaluation Parameters
1. Problem Statement	2	- Clearly defined and relevant to a real-world scenario - Demonstrates innovation or uniqueness - Feasibility within IoT context
2. Introduction & Literature Survey	2	- Clear explanation of the background and need for the project - Review of existing work and justification for the proposed idea

		- Citations/references included
3. Proposed Methodology	3	- Appropriate hardware and software tools selected - Clear and logical block diagram - Working principle well explained - Circuit diagrams/screenshots provided
4. Presentation Skills	1	- Clear and confident delivery - Effective use of slides/visuals - Time management
5. Viva / Q&A	2	- Demonstrates understanding of the project - Able to answer technical questions confidently

➤ **CIE 3 - Project review 3: Development & Testing (25 Marks)**

Includes:

1. Implementation Progress(5),
2. Hardware Assembly & Integration(5)
3. Code Quality, Testing & Debugging(5)
4. Team Collaboration & Involvement(3)
5. Presentation Skills(2)
6. Report(2)
7. Viva(3)

Criteria	Max Marks	Evaluation Parameters
1. Implementation Progress	5	- Completion level of hardware and software modules - Working integration of components - Functionality achieved so far
2. Hardware Assembly & Integration	5	- Neat and safe hardware setup - Proper wiring and connections - Effective use of microcontrollers, sensors, etc.
3. Code Quality, Testing & Debugging	5	- Well-structured, logical, and commented code - Successful testing of modules

		- Debugging done with clear issue-resolution strategy
4. Team Collaboration & Involvement	3	- Equal participation of members - Clear role division and updates - Responsiveness to feedback from the previous review
5. Presentation Skills	2	- Appropriate hardware and software tools selected - Clear and logical block diagram - Working principle well explained - Circuit diagrams/screenshots provided
6. Report	2	- Updated project report with recent progress - Clear documentation of implementation, results, issues, and changes
7. Viva / Q&A	3	- Demonstrates understanding of the project - Able to answer technical questions confidently

Evaluation Rubrics: (SEE-30 marks)

1. Project Functionality & Output (7)
2. System Integration & Innovation (5)
3. Code & Technical Quality (5)
4. Team Collaboration & Involvement(3)
5. Presentation Skills(2)
6. Report & Documentation(5)
7. Viva(3)

Criteria	Max Marks	Evaluation Parameters
1. Project Functionality & Output	7	- Complete working model as proposed - Accurate and reliable output - Real-time data flow and system responsiveness

2. System Integration & Innovation	5	- Seamless integration of hardware and software - Originality or uniqueness of the solution - Usefulness and relevance of the project
3. Code & Technical Quality	5	- Well-written and optimized code - Proper logic implementation - Effective debugging and error handling
4. Team Collaboration & Involvement	3	- Equal participation of members - Clear role division and updates - Responsiveness to feedback from the previous review
5. Presentation Skills	2	- Appropriate hardware and software tools selected - Clear and logical block diagram - Working principle well explained - Circuit diagrams/screenshots provided
6. Report & Documentation	5	- Final report is complete, structured, and well-formatted - Includes circuit diagrams, screenshots, output snapshots, and references
7. Viva / Q&A	3	- Demonstrates understanding of the project - Able to answer technical questions confidently